

My Project

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1 Class Index	1
1 Class Index	1
1.1 Class List	1
2 File Index	1
2.1 File List	1
3 Class Documentation	1
3.1 SudokuBoard Struct Reference	1
3.1.1 Constructor & Destructor Documentation	2
3.1.2 Member Data Documentation	2
4 File Documentation	2
4.1 Sudoku_Algorithm.cpp File Reference	2
4.1.1 Enumeration Type Documentation	3
4.1.2 Function Documentation	3
Index	9

1 Class Index

1.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

SudokuBoard	1
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2 File Index

2.1 File List

Here is a list of all files with brief descriptions:

Sudoku_Algorithm.cpp	2
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3 Class Documentation

3.1 SudokuBoard Struct Reference

Public Member Functions

- [SudokuBoard](#) ()

Public Attributes

- `std::vector< std::vector< int > >` [grid](#)

Static Public Attributes

- static const int [SIZE](#) = 9

3.1.1 Constructor & Destructor Documentation

SudokuBoard()

```
SudokuBoard::SudokuBoard () [inline]
```

3.1.2 Member Data Documentation

grid

```
std::vector<std::vector<int> > SudokuBoard::grid
```

SIZE

```
const int SudokuBoard::SIZE = 9 [static]
```

The documentation for this struct was generated from the following file:

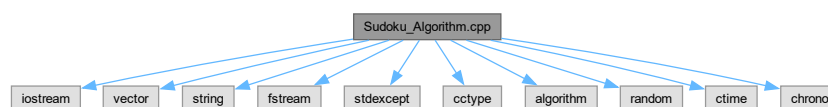
- [Sudoku_Algorithm.cpp](#)

4 File Documentation

4.1 Sudoku_Algorithm.cpp File Reference

```
#include <iostream>
#include <vector>
#include <string>
#include <fstream>
#include <stdexcept>
#include <cctype>
#include <algorithm>
#include <random>
#include <ctime>
#include <chrono>
```

Include dependency graph for Sudoku_Algorithm.cpp:



Classes

- struct [SudokuBoard](#)

Enumerations

- enum class [Difficulty](#) { [EASY](#) , [MEDIUM](#) , [HARD](#) }

Functions

- void [printBoard](#) (const [SudokuBoard](#) &board)
- bool [isValidMove](#) (const [SudokuBoard](#) &board, int row, int col, int num)
- bool [isSolved](#) (const [SudokuBoard](#) &board)
- void [fill3x3Box](#) ([SudokuBoard](#) &board, int startRow, int startCol)
- bool [solveSudoku](#) ([SudokuBoard](#) &board)
- void [generateSudoku](#) ([SudokuBoard](#) &board, [Difficulty](#) level)
- void [playManual](#) ([SudokuBoard](#) &board)
- [SudokuBoard](#) [loadBoard](#) (const std::string &filename)
- void [saveBoard](#) (const [SudokuBoard](#) &board, const std::string &filename)
- int [main](#) ()

4.1.1 Enumeration Type Documentation

Difficulty

```
enum class Difficulty [strong]
```

Enumerator

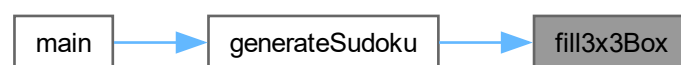
EASY	
MEDIUM	
HARD	

4.1.2 Function Documentation

fill3x3Box()

```
void fill3x3Box (
    SudokuBoard & board,
    int startRow,
    int startCol)
```

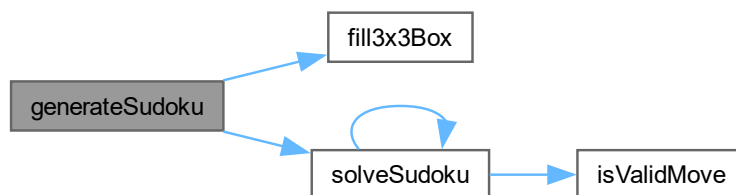
Here is the caller graph for this function:



generateSudoku()

```
void generateSudoku (  
    SudokuBoard & board,  
    Difficulty level)
```

Here is the call graph for this function:



Here is the caller graph for this function:

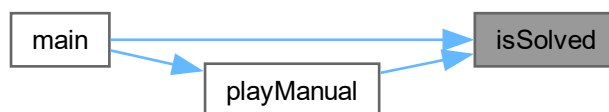
**isSolved()**

```
bool isSolved (  
    const SudokuBoard & board)
```

Here is the call graph for this function:



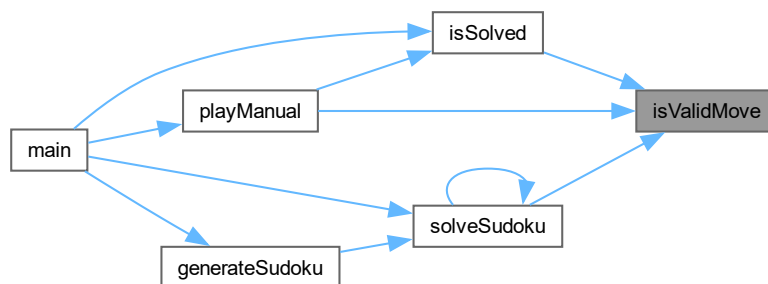
Here is the caller graph for this function:



`isValidMove()`

```
bool isValidMove (  
    const SudokuBoard & board,  
    int row,  
    int col,  
    int num)
```

Here is the caller graph for this function:



`loadBoard()`

```
SudokuBoard loadBoard (  
    const std::string & filename)
```

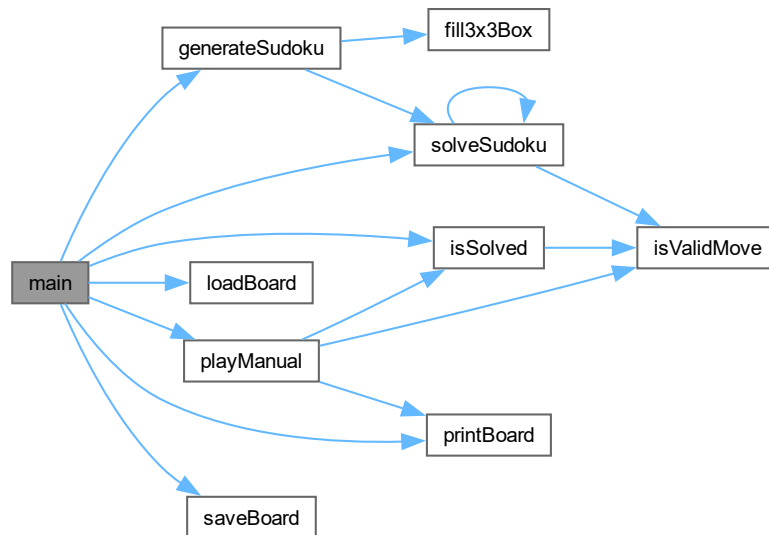
Here is the caller graph for this function:



main()

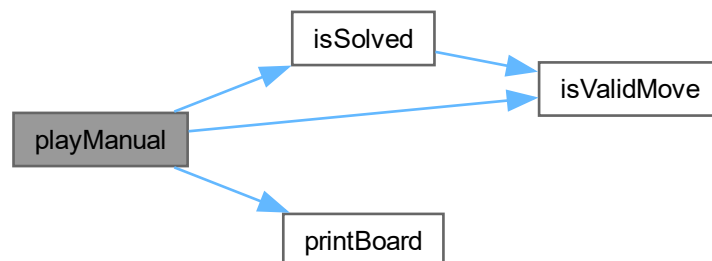
```
int main ()
```

Here is the call graph for this function:

**playManual()**

```
void playManual (  
    SudokuBoard & board)
```

Here is the call graph for this function:



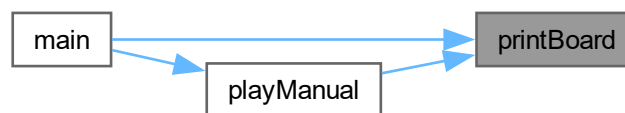
Here is the caller graph for this function:



printBoard()

```
void printBoard (  
    const SudokuBoard & board)
```

Here is the caller graph for this function:



saveBoard()

```
void saveBoard (  
    const SudokuBoard & board,  
    const std::string & filename)
```

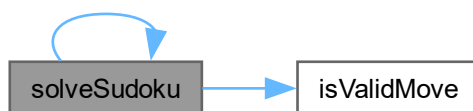
Here is the caller graph for this function:



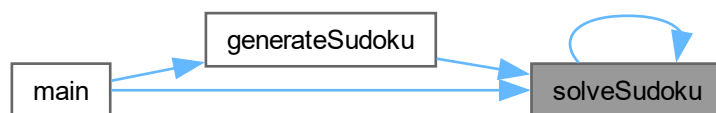
solveSudoku()

```
bool solveSudoku (  
    SudokuBoard & board)
```

Here is the call graph for this function:



Here is the caller graph for this function:



Index

Difficulty
 Sudoku_Algorithm.cpp, [3](#)

EASY
 Sudoku_Algorithm.cpp, [3](#)

fill3x3Box
 Sudoku_Algorithm.cpp, [3](#)

generateSudoku
 Sudoku_Algorithm.cpp, [3](#)

grid
 SudokuBoard, [2](#)

HARD
 Sudoku_Algorithm.cpp, [3](#)

isSolved
 Sudoku_Algorithm.cpp, [4](#)

isValidMove
 Sudoku_Algorithm.cpp, [5](#)

loadBoard
 Sudoku_Algorithm.cpp, [5](#)

main
 Sudoku_Algorithm.cpp, [5](#)

MEDIUM
 Sudoku_Algorithm.cpp, [3](#)

playManual
 Sudoku_Algorithm.cpp, [6](#)

printBoard
 Sudoku_Algorithm.cpp, [7](#)

saveBoard
 Sudoku_Algorithm.cpp, [7](#)

SIZE
 SudokuBoard, [2](#)

solveSudoku
 Sudoku_Algorithm.cpp, [7](#)

Sudoku_Algorithm.cpp, [2](#)

 Difficulty, [3](#)

 EASY, [3](#)

 fill3x3Box, [3](#)

 generateSudoku, [3](#)

 HARD, [3](#)

 isSolved, [4](#)

 isValidMove, [5](#)

 loadBoard, [5](#)

 main, [5](#)

 MEDIUM, [3](#)

 playManual, [6](#)

 printBoard, [7](#)

 saveBoard, [7](#)

 solveSudoku, [7](#)

SudokuBoard, [1](#)

grid, [2](#)
SIZE, [2](#)
SudokuBoard, [2](#)